1. a) Let $\Delta(s)$ and $H(s)$ be stable. The feedback loop is stable if $\|\Delta H\|_\infty < 1$.

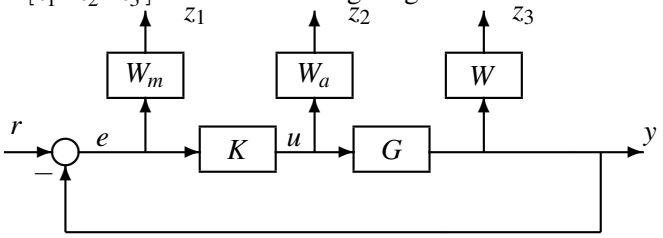
b) For single-input-single-output systems, the condition $\|\Delta H\|_\infty < 1$ implies that the Nyquist diagram of the loop gain $\Delta(s)H(s)$ lies within the unit circle. Since $\Delta(s)$ and $H(s)$ are stable, this implies that the closed loop is stable.

c)
- Suppose that $\Delta_a = 0$ and let the input to Δ_m be ε_2 while the output from Δ_m be δ_2. Then a calculation shows that $\varepsilon_2 = S\delta_2$ where $S = (I+GK)^{-1}$ is the sensitivity. Using the small gain theorem, to satisfy the first design requirement, it is sufficient that $\|\Delta_m(j\omega)S(j\omega)\| < 1, \forall\omega$. This can be satisfied if $\|W_m S\|_\infty < 1$, where $W_m = w_m I$.
- An analogous procedure shows that to satisfy the second design requirement, it is sufficient that $\|\Delta_a(j\omega)K(j\omega)S(j\omega)\| < 1, \forall\omega$. This can be satisfied if $\|W_a KS(j\omega)\|_\infty < 1$, where $W_a = w_a I$.

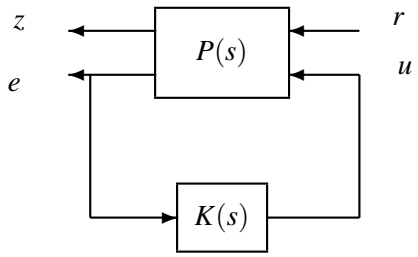
A calculation shows that, when $\Delta_a = 0$ and $\Delta_m = 0$, $y(s) = G(s)K(s)S(s)r(s)$. Thus $\|y(j\omega)\| \le \|G(j\omega)K(j\omega)S(j\omega)\|\|r(j\omega)\|$. It follows that a sufficient condition to achieve the performance design specification is $\|G(j\omega)K(j\omega)S(j\omega)\| < |w^{-1}(j\omega)|, \forall\omega$ or equivalently $\|WGKS\|_\infty < 1$, where $W = wI$.

To satisfy all design requirements, it is sufficient that $\left\| \begin{bmatrix} W_m S \\ W_a KS \\ WGKS \end{bmatrix} \right\|_\infty < 1$.

d) So, the design specifications reduce to the requirement that the transfer matrix from r to $z = [z_1^T \;\; z_2^T \;\; z_3^T]^T$ in the following diagram has $\mathscr{H}_\infty$-norm less than 1.



e) The generalized regulator problem: find an internally stabilizing K such that $\|T_{zr}\|_\infty < 1$



$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{c|c} W_m & -W_m G \\ 0 & W_a \\ 0 & WG \\ \hline I & -G \end{array}\right].$$

 Copyright Imperial College London

2. a) i) A is stable if $A^TP+PA \prec 0$ with $P \succ 0$. To prove this, let $Az=\lambda z$, $z \neq 0$. Then multiplying the inequality from the left by z' and from the right by z gives $(\lambda+\bar{\lambda})z'Pz<0$. Since $P \succ 0$ it follows that $z'Pz>0$ and it follows that $\lambda+\bar{\lambda}<0$ so that A is stable.

ii) Assume that $x(0)=0$. Then $\|H\|_\infty < \gamma$ if and only if $J := \int_0^\infty [y^Ty-\gamma^2u^Tu]dt<0$, for all $u(t)$ such that $\|u\|_2<\infty$. Since $\|u\|_2$ is bounded, $\lim_{t\to\infty} x(t)=0$. Now, $\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}[x^TPx]dt = x(\infty)^TPx(\infty)-x(0)^TPx(0)=0$. So,

$$0=\int_0^\infty \left(\dot{x}^TPx+x^TP\dot{x}\right)dt=\int_0^\infty [x^T(A^TP+PA)x+x^TPBu+u^TB^TPx]dt.$$

Using $y=Cx+Du$ and adding the last expression to J

$$\begin{aligned} J &= \int_0^\infty [x^T(A^TP+PA+C^TC)x+2x^T(PB+C^TD)u+u^T(D^TD-\gamma^2I)u]dt \\ &= \int_0^\infty \begin{bmatrix} x^T & u^T \end{bmatrix} \overbrace{\begin{bmatrix} A^TP+PA+C^TC & PB+C^TD \\ B^TP+D^TC & D^TD-\gamma^2I \end{bmatrix}}^{M} \begin{bmatrix} x \\ u \end{bmatrix} dt. \end{aligned}$$

It follows that $J<0$ if $M \prec 0$. So A is stable and $\|H\|_\infty<\gamma$, if $M \prec 0$ and $P \succ 0$ since $M \prec 0$ also implies that $A^TP+PA \prec 0$.

iii) The dual condition is: A is stable and $\|H\|_\infty<\gamma$ if and only if A^T is stable and $\left\|H^T\right\|_\infty<\gamma$. Using Part a(ii) (replacing (A,B,C,D) by (A^T,C^T,B^T,D^T)), A is stable and $\|H\|_\infty<\gamma$ if there exists $P \succ 0$ such that

$$N := \begin{bmatrix} AP+PA^T+BB^T & PC^T+BD^T \\ CP+DB^T & DD^T-\gamma^2I \end{bmatrix} \prec 0.$$

b) i) Replace u by $u+Fx$, in the state equation gives

$$\dot{x}=(A+BF)x+Bu, \qquad y=Cx.$$

Therefore $T_{yu}(s)=C\left(sI-(A+BF)\right)^{-1}B$.

ii) Using the answer to Part a(iii), replacing A by $A+BF$ and setting $D=0$, there exists a feasible matrix F if there exists $P \succ 0$ such that

$$\begin{bmatrix} (A+BF)P+P(A+BF)^T+BB^T & PC^T \\ CP & -\gamma^2I \end{bmatrix} \prec 0.$$

Since the only nonlinearity is due to the product FP, we let $Z=FP$ and so there exists a feasible gain matrix F if there exists $P \succ 0$ and Z such that

$$\begin{bmatrix} AP+PA^T+BZ+Z^TB^T+BB^T & PC^T \\ CP & -\gamma^2I \end{bmatrix} \prec 0.$$

3. a) Let the realization be partitioned compatibly with P and Q as $G(s) \stackrel{s}{=} \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & 0 \end{array}\right]$.

Then

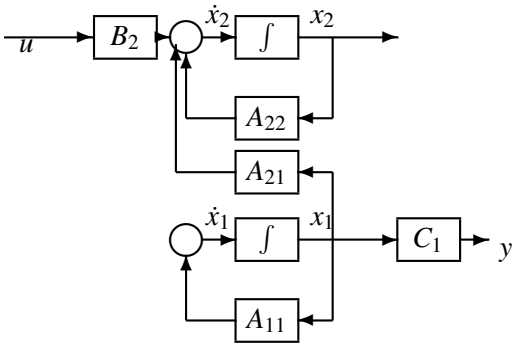
$$AP + PA^T + BB^T = \begin{bmatrix} B_1B_1^T & A_{12}P_2 + B_1B_2^T \\ \star & A_{22}P_2 + P_2A_{22}^T + B_2B_2^T \end{bmatrix} = 0$$

$$A^TQ + QA + C^TC = \begin{bmatrix} A_{11}^TQ_1 + Q_1A_{11} + C_1^TC_1 & Q_1A_{12} + C_1^TC_2 \\ \star & C_2^TC_2 \end{bmatrix} = 0 \quad (3.1)$$

It follows that $B_1 = 0$, $C_2 = 0$ and $A_{12} = 0$. Thus the realization decomposes into two subsystems $G_1 : \dot{x}_1 = A_{11}x_1$ with n_1 modes and $G_2 : \dot{x}_2 = A_{22}x_2 + A_{21}x_1 + B_2u,\ y = C_1x_1$ with n_2 modes. Since A is stable, both sub–systems are stable.

i) G_1 is uncontrollable since $B_1 = 0$. We prove observability by contradiction. Suppose that the pair (A_{11}, C_1) is not observable and hence there exists an eigenvalue λ and an eigenvector $z \neq 0$ such that $A_{11}z = \lambda z$. It follows from the (1,1) entry of (3.1) that $(\lambda + \bar{\lambda})z'Q_1z + z'C_1^TC_1z = 0$. If $C_1z = 0$ then $\lambda + \bar{\lambda} = 0$ since $Q_1 \succ 0$, which contradicts the assumption that A is stable. Thus the realization is observable.

ii) For the subsystem G_2 it is clear that all the modes are unobservable and controllable using a dual argument to the above.



b) i) Using the result that A is stable if and only if there exists $P = P^T \succ 0$ such that $A^TP + PA \prec 0$ we have $\mathscr{S}$ is simultaneously detectable if and only if there exist $L \in \mathscr{R}^{n\times m}$ and $P_i = P_i^T \succ 0$ such that $(A_i + LC_i)^TP_i + P_i(A_i + LC_i) \prec 0$, $i = 1, \ldots, N$.

ii) The above conditions are nonlinear, but can be linearized by taking $P_i = P$ for all i, and defining a variable $\hat{L} := PL$ and so a set of sufficient conditions is: $\mathscr{S}$ is simultaneously detectable if there exist $\hat{L} \in \mathscr{R}^{n\times m}$ and $P = P^T \succ 0$ such that $A_i^TP + PA_i + \hat{L}C_i + C_i^T\hat{L}^T \prec 0$, $i = 1, \ldots, N$. If the conditions are satisfied, then $L = P^{-1}\hat{L}$.

4. a) The generalized regulator set–up is given by

$$\begin{bmatrix} z(s) \\ y(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u(s) \end{bmatrix},\ u(s) = Fy(s), P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c||c|c} A & B & B \\ \hline\hline C & 0 & 0 \\ \hline 0 & 0 & I \\ \hline\hline I & 0 & 0 \end{array}\right].$$

b) Let the cost function be defined as $J = \|z\|_2^2 - \gamma^2 \|w\|_2^2$. Let $V = x^T X x$ and set $u = Fx$. Provided that $X = X^T \succ 0$ and $\dot{V} < 0$ along the closed loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. It follows that

$$\dot{V} = \dot{x}^T X x + x^T X \dot{x} = x^T \left(A^T X + XA + F^T B^T X + XBF\right) x + x^T XBw_1 + w_1^T B^T Xx.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$-x_0^T X x_0 = \int_0^\infty [x^T \left(A^T X + XA + F^T B^T X + XBF\right) x + x^T XBw_1 + w_1^T B^T Xx] dt.$$

Using the definition of J, adding the last equation, and completing the squares,

$$J = x_0^T X x_0 + \int_0^\infty \{x^T \left[A^T X + XA + C^T C - X\left(BB^T - \gamma^{-2} BB^T\right) X\right] x + \|Zx\|^2 - \|v\|^2\} dt$$

where $Z = F + B^T X$ and $v = \gamma w - \gamma^{-1} B^T X x$.

c) The specification $\|H\|_\infty \le \gamma$ is equivalent to $J \le 0$ when $x_0 = 0$. Therefore sufficient conditions for $J \le 0$ are $Z = 0$ and the existence of a matrix X such that

$$A^T X + XA + C^T C - X\left(BB^T - \gamma^{-2} BB^T\right) X = 0, \qquad X = X^T \succ 0.$$

Setting $Z = 0$ gives $F = -B^T X$. The worst case disturbance is obtained by setting $v = 0$, so $w^* = \gamma^{-2} B^T X$. The closed loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_c x$ where $A_c = A - \left(BB^T - \gamma^{-2} BB^T\right)$ and a further condition is $Re\ \lambda_i(A_c) < 0,\ \forall\ i$.

d) Using the expression for J, with $w = 0$ but with x_0 not necessarily zero, and the solution in Part c) with $\gamma \to \infty$, it follows that the minimum value of J is $J = x_0^T X x_0$ achieved by $F = -B^T X$ and the Riccati equation becomes

$$A^T X + XA + C^T C - XBB^T X = 0, \qquad X = X^T \succ 0.$$

Thus, as $\gamma \to \infty$, the solution of the $\mathscr{H}_\infty$ regulator converges to that of the LQR.

e) Using $u = Fx$ in the state equations gives

$$\begin{bmatrix} \dot{x} \\ z \end{bmatrix} = \left[\begin{array}{c|c} A + BF & B \\ \hline C & 0 \\ F & 0 \end{array}\right] \begin{bmatrix} x \\ w \end{bmatrix}.$$

Using the Bounded Real Lemma from part a(iii) of question 2, with the substitution $Y = FP$ gives the LMI sufficient conditions

$$\begin{bmatrix} AP + PA^T + BY + Y^T B^T + BB^T & PC^T & Y^T \\ CP & -\gamma^2 I & 0 \\ Y & 0 & -\gamma^2 I \end{bmatrix} \prec 0, \quad P \succ 0.$$